

8TH GRADE MATH • SUPPORT NOTES

Unit 3: Equations & Inequalities

NC.8.EE.7

Name: _____

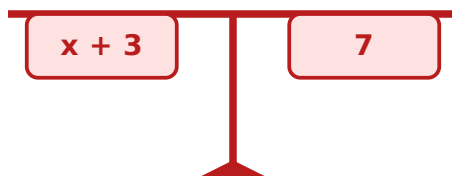
Date: _____

Class: _____

Period: _____

★ The One Big Idea

Solving an equation means **getting the variable alone**. To undo something, do the **opposite** — and do it to **both sides**.



Both sides must stay equal — do the same to BOTH sides



Warm-Up: Inverse (Opposite) Operations

Each operation has an opposite

To undo...	Do the opposite
+ adding	– subtract
– subtracting	+ add
× multiplying	÷ divide
÷ dividing	× multiply

Words to Know

Variable — a letter that stands for an unknown number

Coefficient — the number multiplied by the variable (in $3x$, the 3)

Like terms — terms with the same variable: $3x$ and $5x$

Solution — the value that makes the equation true

3.1 & 3.2 One-Step & Two-Step Equations

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 One-Step: $x + 5 = 12$

Undo +5 by -5 on both sides: $x + 5 - 5 = 12 - 5 \rightarrow x = 7$

 Two-Step Order (undo backwards)

1. Undo $+$ or $-$ first.
2. Undo $\times$ or $\div$ second.

Example: $2x + 3 = 11$

Step 1: -3 both sides $\rightarrow 2x = \underline{\hspace{2cm}}$

Step 2: $\div 2$ both sides $\rightarrow x = \underline{\hspace{2cm}}$

 You Try

$3x - 4 = 11 \rightarrow x = \underline{\hspace{2cm}}$

3.3 Distributive Property

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★ **Multiply the outside by EACH inside term**

$$a(b + c) = a \times b + a \times c$$

$$3(x + 4) = 3x + 12$$

🔗 **I Do: $2(x + 5) = 16$**

Distribute: $2x + 10 = 16$

-10 : $2x = 6 \rightarrow \div 2$: **$x = 3$**

⚠ **Watch Out**

Do not forget the second term! $3(x+4)$ is NOT $3x+4$. It is $3x+12$.

✍ **You Try**

$$4(x - 2) = 20 \rightarrow x = \underline{\hspace{2cm}}$$

3.4 & 3.5 Variables on Both Sides / Multi-Step

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 The 5-Step Plan

1. **Distribute** if there are parentheses.
2. **Combine** like terms on each side.
3. **Move variables** to one side (add/subtract).
4. **Move numbers** to the other side.
5. **Divide** to get the variable alone.

 I Do: $5x + 2 = 3x + 10$

Subtract $3x$: $2x + 2 = 10$

Subtract 2: $2x = 8$

Divide 2: **$x = 4$**

 You Try

$$6x - 1 = 2x + 11 \rightarrow x = \underline{\hspace{2cm}}$$

3.6 Special Cases

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★ Two Strange Endings

No Solution: variables disappear and you get a **false** statement, like $5 = 8$.

Infinite Solutions: variables disappear and you get a **true** statement, like $6 = 6$.

One Solution: you get $x = \text{a number}$ (the normal case).

👤 I Do

$2x + 4 = 2x + 9 \rightarrow \text{subtract } 2x \rightarrow 4 = 9 \text{ (false)} \rightarrow \text{No Solution}$

$3(x+2) = 3x + 6 \rightarrow 3x+6 = 3x+6 \rightarrow 6 = 6 \text{ (true)} \rightarrow \text{Infinite Solutions}$

3.7 Linear Inequalities

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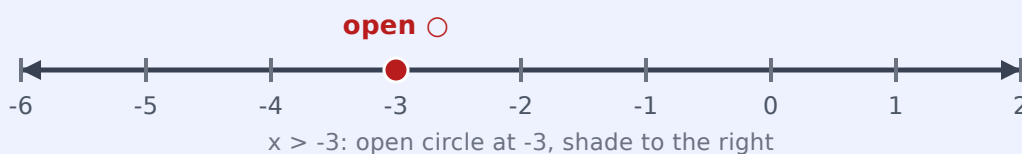
★ Solve like an equation — with ONE special rule

Symbols: $<$ less than, $>$ greater than, $\leq$ at most, $\geq$ at least.

FLIP the symbol when you multiply or divide both sides by a **negative** number.

👤 I Do: $-2x < 6$

Divide by -2 (flip!): $x > -3$

**⚠ Open vs. Closed Circle**

$<$ or $>$ → **open** circle (not included). $\leq$ or $\geq$ → **closed** circle (included).

3.8 Word Problems

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☰ Turn Words Into an Equation

Words	Math
sum, more than, increased by	+
difference, less than, fewer	-
product, times, of	×
is, equals, results in	=

👤 I Do

“5 more than twice a number is 17.”

$2x + 5 = 17 \rightarrow 2x = 12 \rightarrow \mathbf{x = 6}$

Part 2 Practice **Solve each equation (show both sides)**

#	Equation	x =
1	$x - 7 = 10$	
2	$3x = 18$	
3	$2x + 5 = 19$	
4	$4(x + 1) = 24$	
5	$5x - 2 = 3x + 8$	
6	$2x + 6 = 2x + 1$ (special?)	

 **Inequalities**

1. $x + 4 > 9 \rightarrow$

2. $-3x \leq 12 \rightarrow$ (flip?)

★ ANSWER KEY ★ **Guided Notes**

3.2: $2x=8$, $x=4$; You try $x=5$. 3.3: $x=8$. 3.4/3.5: $x=3$. 3.6: covered above.

3.7: $-2x < 6 \rightarrow x > -3$. 3.8: $x=6$.

 Practice

1) $x=17$ 2) $x=6$ 3) $x=7$ 4) $x=5$ 5) $x=5$ 6) No solution ($6 \neq 1$)

Ineq: 1) $x > 5$ 2) $x \geq -4$ (flip: divide by -3)

End of Unit 3 support packet.